

SHIVAM SINGH

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SENIOR PRODUCT MANAGER · GENERATIVE & AGENTIC AI · COPILOTS, LLM PLATFORMS & RESPONSIBLE AI

EDUCATION

University of Michigan, Stephen M. Ross School of Business — Master of Business Administration (MBA), 2022–2024 · Exchange semester: London School of Economics (LSE), Economics, Markets & Public Policy (2024)

MNNIT Allahabad — Bachelor of Technology (B.Tech), Computer Science & Engineering

Peer-reviewed publication, Journal of Cleaner Production (Elsevier) — 159 citations · Published author

PROFESSIONAL EXPERIENCE

Amazon (AWS) — Senior GenAI Strategist (Product Strategy & Roadmap Owner)

2024 – Present

Agentic GenAI shopping assistant (Rufus) · Consumer surface with 100M+ users

- Drove product strategy and roadmap for Rufus, an agentic GenAI conversational shopping assistant at 100M+ user scale: pivoted from open-ended chat to solving decision paralysis in purchase journeys, partnering with engineering, UX research, applied science, and international teams to rebuild it as an interactive commerce experience; owned the full product lifecycle (concept→beta→GA) and post-launch iteration on telemetry and model-quality signals, defining the Q-Score LLM evaluation metric as the ship/no-ship release bar — cut hallucination rate to ~2.8%, lifted session depth +15%, and drove ~25% adoption.
- Led product for secure agentic AI workflows for regulated enterprises: shipped a fully managed platform enabling GenAI applications to retrieve private enterprise data (RAG) and execute business tasks via tool use, prioritizing transparency and auditability of agent reasoning (chain-of-thought exposure) over raw latency to unlock adoption in financial services; drove intelligent model routing (lightweight models for simple queries, frontier models for complex reasoning) to optimize customers' inference cost-to-serve — hands-on ownership of agent orchestration, LLM evaluation, prompt engineering, and reliability tradeoffs.
- Built responsible AI guardrails for GenAI ad creative (image/video) generation: re-architected the pipeline to evaluate content safety and regulatory compliance policy pre-generation rather than post-hoc moderation, translating advertising regulations into deterministic policy-as-code with legal, platform engineering, and trust-and-safety teams — slashed creative rejection rates to near zero and compressed campaign launch time from days to minutes, converting a moderation bottleneck into a growth lever.
- Led a \$50M+ buy-vs-build M&A initiative (agentic OS acquisition) to expand the portfolio from copilots to autonomous agents: uncovered unscalable GPU compute unit economics in commercial due diligence, modeled a TCO restructuring on proprietary custom silicon to fix cost-to-serve pre-close, and directed adversarial red-team stress tests of agent safety and alignment protocols across Corporate Development, engineering, and legal.
- Shipped Japan LLM localization for the global recommendation engine: halted a US-centric rollout after identifying cultural and linguistic failure modes, re-architected the ML platform to balance horizontal scalability with locale intelligence (SentencePiece tokenization, Keigo honorific handling, culturally-aware ranking) — established the reusable internationalization (i18n/l10n) playbook for global market expansion.

Rakuten — MBA Product Intern, AI Platform & Merchant Solutions

2023 · Tokyo, Japan

- Audited unit economics of a planned direct-to-consumer international launch for the cashback/merchant platform, stress-tested CAC and LTV assumptions, proved the model too fragile, and steered leadership to a partnership-led B2B2C acquisition strategy: built segmentation and data infrastructure to identify highest-LTV user cohorts, enabling large-scale corporate partnerships over volatile paid-media channels.
- Drove cross-border commerce go-to-market (US-Japan corridor) and championed automated merchant KYC verification and self-serve onboarding, cutting time-to-first-transaction to accelerate merchant activation, marketplace liquidity, and market-share growth.

The D. E. Shaw Group — Platform Lead, Trading Infrastructure (Internal Product Owner)

2016 – 2020 · Hyderabad, India

- Owned the end-to-end operational lifecycle of a global high-performance computing (HPC) fleet powering automated trading: closed the gap between hardware capex and P&L impact by standing up the capacity planning, validation, observability, and performance-optimization program with hardware engineering and finance leadership — reduced infrastructure OpEx by \$1.2M.
- Product lead for an Intelligent Document Processing (IDP) platform for investment infrastructure: replaced manual data entry and fragmented spreadsheets with an ML-driven, confidence-scored, human-in-the-loop extraction pipeline with active-learning feedback, aligning software engineers, legal counsel, and traders — collapsed earnings-report ingestion from hours to seconds and became the deal-operations backbone for the firm's venture strategy.
- Drove a real-time streaming risk engine replacing end-of-day batch reporting: de-risked cutover with a 3-month "shadow mode" parallel-run validation against live production transactions, cutting false-positive alerts ~90%, achieving millisecond-latency pre-trade enforcement, and unlocking \$85M in previously reserved regulatory capital.

McKinsey & Company — Associate, Digital Product (Fintech)

2014 – 2016 · Gurgaon, India

- Led the 0→1 launch of a Southeast Asia digital neobank for the unbanked: validated an alternative-data (telco) credit-scoring model with financial regulators in a market where 70% lacked formal credit history, and hit 100K active users in 90 days at sub-benchmark CAC (vs. ~\$175 industry acquisition cost) with loan-approval uplift from 10% to 40%.
- Built a "Next Best Action" ML churn-prediction engine for wealth-management advisors: pivoted from opaque black-box risk scores to explainable AI with plain-English recommendations and an advisor-override feedback loop, solving the trust and adoption barrier — then rapidly retrained the model when the 2020 market dislocation shifted client behavior from dissatisfaction to fear, proving model resilience under regime change.
- Designed a dynamic pricing engine for a major retail client: segmented traffic-driving key-value items from margin-driving long-tail inventory, won executive buy-in by backtesting the model against historical sales with hard guardrails on price volatility, and re-engineered the pricing governance workflow to cut update cycles from months to days.